Control Tower (CT-01…CT-07) — Architecture Audit & Blueprint

Version 1.0 · Current-state audit of 7 n8n workflows · Diagrams delivered as separate editable files (draw.io, SVG, PNG, Mermaid)

Evidence basis. Everything below is derived from the uploaded workflow JSON only. Nothing was executed; no credential, API, table or runtime behaviour was verified. Each statement is tagged **Confirmed** (visible in the JSON), **Inferred** (reasoned from the JSON) or **Verify** (cannot be established from the files).

1. Executive summary

The system is an event-driven *Business Operations Control Tower* built from seven n8n workflows. Business systems POST events to CT-01. Each event is validated, hashed, de-duplicated, stored as an **incident**, assessed by a Gemini-based AI agent (with optional HubSpot customer context), scored by deterministic code, optionally sent to a human approver by email, and finally dispatched as a Slack alert (CT-03). Every state change goes through a single writer (CT-02), which also appends an audit row. CT-05 (scheduled) retries failed dispatches, escalates stale incidents and detects systemic failure; CT-06 lets a human resume ESCALATED/FAILED incidents; CT-04 is a global error handler; CT-07 would write outcomes back to HubSpot but is **disabled** in the export.

Strengths (Confirmed). One state-machine writer with an explicit legal-transition table; an audit row per transition; deterministic scoring and approval rules (the AI cannot approve); a safe AI fallback that forces human approval; immediate 202 response before slow work; retry loop with linear back-off; stale-incident and systemic-failure detection.

Headline risks.

1. Both inbound webhooks have **no authentication configured** (F1, Critical), and the recovery endpoint can resume an incident whose approval timed out without any approval being recorded (F2, High).
2. The global error handler is attached to **CT-06 only**, not to CT-01 or CT-05 as its own note states (F3, High).
3. The AI prompt and the structured-output schema **contradict each other** (F4, High); the likely effect is frequent AI fallback and mandatory human approval.
4. State transitions are **not concurrency-safe or atomic** (F5, High).

5. The HubSpot write-back (CT-07) is built but **never invoked** (F6, Medium).

The architecture itself is sound and does not need to grow. The recommended target state keeps all seven workflows and adds: webhook authentication, an error workflow on every workflow, a compare-and-swap state write, an aligned AI contract, approval integrity on recovery, DB-level idempotency, and two small tables only if justified (Section 8).

2. Inventory of uploaded workflows

Files received: 9 JSON files + 1 screenshot, covering **7 distinct workflows**. Two workflows were uploaded twice (CT-01: versions "(4)" and "(6)"; CT-07: an older inactive version and " (1)"). This audit uses the latest of each: CT-01 (6), CT-02 (3), CT-03 (2), CT-04 (1), CT-05 (2), CT-06 (1), CT-07 (1). The older CT-07 used the native HubSpot Engagement operation for notes; the latest uses an HTTP Request node.

ID	Workflow	Status	Trigger	Functional nodes	Purpose
CT-01	Main — Event Intake & Orchestration	active	Webhook POST /ops-events	52 (incl. AI sub-nodes)	Validate, sign, de-duplicate, create incident, CRM context, AI assessment, score & route, approval, dispatch, first retry scheduling
CT-02	Sub — State & Audit Writer	active	Execute Workflow trigger	9	Only writer of incidents.state: validates transition, updates incident, writes audit_logs; hook to CT-07 (disabled)
CT-03	Sub — Dispatch Action	active	Execute Workflow trigger	5	Reads system_config, builds Slack message, posts to Slack (3 tries); fails on purpose if Slack fails
CT-04	Global Error Handler	active	Error Trigger	8	Slack alert; mark open incident FAILED via CT-02, otherwise audit row only
CT-05	Watchdog	active	Schedule (5 min)	24	A: retry loop · B: stale scan · C: systemic-failure detection
CT-06	Human Recovery Endpoint	active	Webhook POST /ops-recover	7	ESCALATED/FAILED → RETRYING (retry_count 0) via CT-02

ID	Workflow	Status	Trigger	Functional nodes	Purpose
CT-07	Sub — HubSpot Outcome Sync	active, **never called**	Execute Workflow trigger	9	Find HubSpot contact by email, add a note with the outcome, audit the result

Workflow-to-workflow relationships (Confirmed from workflowId references):

- CT-01 → CT-02 (11 call sites) and → CT-03 (error output routes to RETRYING).
- CT-05 → CT-02 (4) and → CT-03.
- CT-06 → CT-02; CT-04 → CT-02.
- CT-02 → CT-07 (fire-and-forget, **node disabled**).
- CT-04 is referenced as Error Workflow only by CT-06.
- No unreachable nodes were found in any workflow (AI model, memory and parser nodes are attached to the agent).

Incident state machine (Confirmed, CT-02 table). RECEIVED → VALIDATING → AI_PROCESSING → CLASSIFIED → (AWAITING_APPROVAL → APPROVED) → PROCESSING → COMPLETED. Side exits: REJECTED (final), RETRYING (from PROCESSING), ESCALATED and FAILED (from any open state; both recoverable to RETRYING). COMPLETED and REJECTED are final.

Key decision rules (Confirmed, CT-01 code). Risk = (severity weight × 15 + AI impact × 4) × department weight, capped at 100; priority P1 ⩾ 80, P2 ⩾ 60, P3 ⩾ 35, else P4; SLA 30 / 120 / 480 / 1440 min. Approval is required if severity is HIGH/CRITICAL, risk ⩾ 75, AI failed, or AI confidence < 0.60. Approval waits 45 min; no answer ⇒ ESCALATED. Dispatch retries: CT-01 failure sets retry_count 1; each Watchdog failure adds 1; at max_retries (default 3) ⇒ ESCALATED with email and Slack.

3. Current-state architecture (VIEW 1)

See **view1_executive** (SVG/PNG/draw.io page 1). Layers: business systems and triggers → n8n orchestration → AI processing and decision logic → external integrations (Slack, Gmail, HubSpot, Gemini) → data layer (Supabase, Postgres chat memory) → human governance and outcomes, with a cross-cutting reliability/security/observability band. Red tags mark audit findings; dashed grey marks the disabled CT-07 path.

4. Detailed data-flow and decision diagrams (VIEW 2A, 2B)

See **view2a_intake_ai_approval** (intake, CRM context, AI, scoring, approval) and **view2b_dispatch_retry_recovery** (dispatch, retry loop, stale scan, systemic alert, human recovery, error handling). Every decision diamond uses conditions read from the node expressions; proposed links are dashed teal.

5. Database and ERD blueprint (VIEW 3)

No SQL schema was supplied. The tables below are **evidenced by Supabase node operations**; column names come from field mappings and expressions. Types, keys and foreign keys are **not established** (shown as inferred).

Table	Purpose	Reads	Inserts	Updates	Owner of writes
incidents	One row per event: raw fields, AI assessment, scoring, state, retry counters	CT-01 (dup check), CT-04, CT-05 ×3, CT-06, CT-07	CT-01	CT-01 (classification), CT-02 (state + patch)	CT-02 for state
audit_logs	Append-only transition and event log	—	CT-02, CT-04, CT-05, CT-07	—	CT-02
approvals	Approval request and decision	—	CT-01	CT-01	CT-01
system_config	Dispatch settings (dispatch_channel_id, simulate_slack_failure)	CT-03	— (no writer present)	—	Manual / external
chat-memory table (Postgres)	AI conversation history, session key ct-hs-<contactId> or ct-ent-<dept>-<entity>	CT-01 agent	CT-01 agent	—	Node-managed (table name = node default, Inferred)

Relationships (Inferred): audit_logs.incident_id → incidents.id and approvals.incident_id → incidents.id (1:N each). audit_logs rows written by CT-04 (error-only) and CT-05 (systemic alert) omit incident_id, so that column must be nullable.

Integrity observations (Confirmed from workflow logic). De-duplication is SELECT-then-INSERT with no unique constraint evidenced. CT-02 updates incidents WHERE id only, then inserts the audit row in a second call, with no transaction. CT-01 writes classification fields directly (outside CT-02).

Recommended structures (proposed, not present): columns/constraints on incidents (UNIQUE(event_id), version, claimed_by/claimed_at, indexes on state+next_retry_at and execution_id); approvals.decided_by and decision_source; crm_sync_log (only if CT-07 is enabled); dead_letter_events (Phase 4). The brief's other candidate tables (business_events, assessment_results, incident_records, retry_records, integration_logs) are **not recommended**: incidents already holds the event, the AI assessment and retry counters, and audit_logs records every transition.

6. API and integration register (VIEW 4)

Native = built-in n8n node. Credential values are never shown; only credential types are named.

Service	Calling workflow · node	Operation	Auth	Error / retry behaviour	Concerns and recommendation
Supabase (native)	All workflows · 17 DB nodes	getAll / insert / update on incidents, audit_logs, approvals, system_config	supabaseApi credential; key privilege not visible (Verify)	Only CT-04 lookup continues on error; no retries on any DB node	Use least-privilege key; add retry on reads; move state+audit into one transactional call
Slack (native)	CT-01 ×2, CT-03, CT-04, CT-05 ×3	Post message to channel	slackApi	CT-03 retry ×3 (2 s); CT-04 continues on error; others none	Channel IDs hard-coded in 6 nodes while CT-03 reads config; externalise; add error handling in CT-05

Service	Calling workflow · node	Operation	Auth	Error / retry behaviour	Concerns and recommendation
Gmail (native)	CT-01 Request Approval; CT-05 Escalation Email	sendAndWait (45 min, approve/reject); send	gmailOAuth2	CT-01 continues on error (becomes TIMEOUT); CT-05 none	Recipient hard-coded and inconsistent; separate EMAIL_FAILED from TIMEOUT; quota not assessed
Google Gemini (native LangChain)	CT-01 AI Agent chat model	Chat completion, temperature 0.2	googlePalmApi	Retry ×2, then deterministic fallback	Model name not pinned in export (Verify); schema conflict (F4)
Postgres (native)	CT-01 Postgres Chat Memory	Session history read/write	postgres credential (separate from Supabase)	None	Retention and PII of memory undefined (F14)
HubSpot (native)	CT-01 and CT-07 Find Contact	Contact search: email EQ, limit, sort by createdate	hubspotAppToken	Retry ×3, continue on error	Customer email used as query; scopes Verify
HubSpot (HTTP Request)	CT-07 Create Note	POST /crm/v3/objects/notes, association to contact	predefined credential type hubspotAppToken	No retry (deliberate, avoids duplicate notes); continue on error	Never invoked. Native Engagement operation does not appear to support notes (Verify) so do not replace without confirming
Inbound webhooks	CT-01 /ops-events, CT-06 /ops-recover	Receive JSON, respond via Respond node	**None configured**	Validation in code only	F1

7. Findings and risks

Severity reflects business impact if the issue is real. **Confirmed** items are visible in the JSON; runtime effects are labelled.

ID	Severity	Finding	Evidence	Impact	Fix	Changes architecture?
F1	**Critical**	No authentication on POST /ops-events or POST /ops-recover	Webhook options are empty in CT-01 and CT-06 (Confirmed); CT-01 note says "Authenticate"	Anyone with the URL can inject events (AI cost, Slack noise, CRM lookups) or resume incidents with a self-asserted actor	Header-auth or HMAC credential on both webhooks; verified actor; rate limit at the edge	Adds an auth gate (no new workflow)
F2	High	Recovery bypasses the approval gate	Approval timeout ⇒ ESCALATED (CT-01); CT-06 accepts any ESCALATED incident and sets RETRYING; CT-05 then dispatches. No check of approvals.status (Confirmed)	An unapproved HIGH/CRITICAL incident can be dispatched; with F1 it is unauthenticated	For approval-timeout incidents, recovery must re-request approval or require a recorded APPROVED decision	Small logic change in CT-06
F3	High	Error handler not attached to CT-01 and CT-05	errorWorkflow set only in CT-06 settings; CT-04 note says CT-01, CT-05, CT-06 (Confirmed). Same for CT-02, 03, 07	Failures in intake or the Watchdog raise no alert and mark no incident FAILED; a failed Watchdog is undetected	Set CT-04 as Error Workflow on every workflow	No
F4	High	AI prompt and output schema contradict	Prompt demands 13 fields incl. recommended_actions[], business_impact, approval/incident flags, one-word category; schema requires category enum (REVENUE_RISK…), recommended_action (string); prompt defines LOW/MEDIUM/HIGH but validator allows CRITICAL (Confirmed)	Parser validation likely fails ⇒ retries ⇒ AI_FALLBACK ⇒ forced approval and generic summary (Inferred, not run)	One schema as the source of truth; align prompt, parser and the code that reads it; add CRITICAL; replay sample events	No
F5	High	State writes are not concurrency-safe or atomic	Transition validated against caller-supplied previous_state; UPDATE filters on id only; audit insert is a separate call (Confirmed)	Concurrent actors can overwrite state; audit row can be lost after a successful update	UPDATE … WHERE id AND state = previous_state (RPC), fail on 0 rows; write incident + audit in one DB function	Small change inside CT-02

ID	Severity	Finding	Evidence	Impact	Fix	Changes architecture?
F6	Medium	HubSpot write-back inactive	CT-02 hook node disabled; CT-07 has no other caller (Confirmed)	CRM never receives outcomes; notes in CT-02/CT-07 describe it as live. If enabled: one note per final-state transition and CT-07 reads current, not triggering, state	Enable only with idempotency (crm_sync_log) or retire CT-07	Optional table
F7	Medium	De-duplication is racy; payload_hash unused	SELECT then INSERT; no unique constraint evidenced; hash stored only (Confirmed)	Simultaneous identical events can create two incidents; same event_id with different body is silently treated as duplicate	UNIQUE(event_id), conflict handling; compare payload_hash	DB constraint
F8	Medium	Approver and channel settings are placeholders / hard-coded	approvals.approver_email and the actor constant use a placeholder; Gmail sendTo uses another address; escalation email text contains a placeholder; Slack channel IDs hard-coded in 6 nodes (Confirmed)	Audit trail names an approver who may not be the recipient; configuration drift	Move to system_config; record the real approver identity	No
F9	Medium	Approval timing and failure semantics	P1 SLA 30 min but approval wait 45 min; Gmail failure continues and is recorded as "no decision within 45 minutes" (Confirmed)	Critical incidents wait longer than their SLA; email outage looks like a human timeout	Shorter wait for P1 plus parallel Slack page; distinct EMAIL_FAILED outcome	No
F10	Medium	Watchdog notifications unprotected	CT-05 Gmail/Slack nodes have no retry or onError and no Error Workflow (Confirmed)	One failed email/Slack call aborts the loop for remaining incidents, silently	continueOnFail with logging; Error Workflow	No
F11	Medium	Overlapping Watchdog runs can double-dispatch	No claim/lock before RETRYING → PROCESSING (Confirmed); overlap Inferred	Same incident dispatched twice	Claim with compare-and-swap (F5) or lease columns	Small
F12	Medium	Error correlation misses non-CT-01 failures	CT-04 finds incident by execution_id; incidents.execution_id is set by CT-01 only (Confirmed)	Failures in CT-05/06 are audit-only	Pass incident_no into errors, or log workflow-level incidents separately	No

ID	Severity	Finding	Evidence	Impact	Fix	Changes architecture?
F14	Medium	Customer data in memory and CRM	Customer email in HubSpot search; chat memory keyed by contact/entity retains event JSON; AI summary goes into Slack/HubSpot notes (Confirmed config; retention Verify)	Privacy and cross-incident context contamination	Define retention, limit memory window, avoid raw payload in memory	No
F13	Low	Demo settings in production path	simulate_slack_failure switch; retry delay 1 min "demo value"; schedule note (Confirmed)	A config row can force dispatch failure	Remove switch or guard by environment	No
F15	Low	Slack dispatch not idempotent	CT-03 retry ×3 and CT-05 re-dispatch	Duplicate alerts on lost responses	Idempotency key in message footer / dedupe	No
F16	Low	Implicit filter semantics	CT-03 and CT-05 list several equality filters on one column without matchType (node default assumed to be "any") (Verify)	Wrong results if default differs	Set matchType explicitly	No
F17	Low	Credential sharing, no tests	Same credentials reused everywhere; pinData empty in all files	Over-privilege; no regression safety net	Per-purpose credentials; scripted test events	No
F18	Low	AI input is untrusted text	Payload enters prompt and memory; output reaches approvers; no tools attached; severity is authoritative (Confirmed)	Limited blast radius; misleading summary possible	Keep no-tools design; sanitise output shown to approvers	No

8. Recommended target-state architecture (VIEW 5)

Same seven workflows. Changes (P1…P8 in the diagram): **P1** authenticate both webhooks; **P2** Error Workflow on all workflows; **P3** compare-and-swap, transactional state+audit write in CT-02; **P4** aligned AI contract with validation; **P5** approval integrity on recovery; **P6** enable CT-07 with idempotency or retire it; **P7** DB unique event_id, claim/lease for the Watchdog, dispatch idempotency; **P8** externalise approver, channel IDs and retry delays into system_config. Design decision: no new orchestration layer or queue is introduced; the existing single-writer pattern is kept because it already gives one audit point.

9. Prioritised implementation roadmap

Phase	Item	Priority	Rationale	Workflow	Depends on	Benefit
1 Critical defects & security	Authenticate /ops-events and /ops-recover (F1)	P0	Open endpoints	CT-01, CT-06	n8n credential	Closes injection and spoofing
1	Approval check on recovery (F2)	P0	Gate can be skipped	CT-06	F1	Governance restored
1	Error Workflow on all workflows (F3)	P0	Silent failures	All	—	Alerts and FAILED states
1	Align AI prompt, schema and parser (F4)	P0	AI likely always falls back	CT-01	Sample events	Working AI assessment
2 Reliability & integrity	Compare-and-swap + atomic audit (F5, F11)	P1	Race and lost audit	CT-02, CT-05	DB function	Consistent state
2	UNIQUE(event_id), payload_hash check (F7)	P1	Duplicate incidents	CT-01 + DB	Data cleanup	Idempotent ingest
2	Protect Watchdog notifications (F10)	P1	Loop aborts silently	CT-05	F3	Loop completes
2	Approval timing and EMAIL_FAILED (F9)	P2	SLA mismatch	CT-01	—	Accurate outcomes
3 Optimisation & maintainability	Externalise config (F8, F13, F16)	P2	Drift and demo values	CT-01, 03, 05	system_config	Single source of config
3	Decide CT-07: enable with crm_sync_log or retire (F6)	P2	Dead feature	CT-02, CT-07	HubSpot scopes	CRM visibility
3	Memory retention and privacy rules (F14, F18)	P2	PII	CT-01	Policy	Compliance
4 Scale & monitoring	Error correlation by incident_no (F12)	P3	Missed linkage	CT-04	—	Better recovery
4	Dispatch idempotency (F15) and dead_letter_events	P3	Duplicates, lost events	CT-03, CT-01	—	Replay and de-dupe

Phase	Item	Priority	Rationale	Workflow	Depends on	Benefit
4	Scripted test events, metrics (F17)	P3	No tests	All	—	Regression safety

10. Testing and validation checklist

- Post an event without credentials to each webhook: expect rejection (after P1).
- Valid, invalid, and duplicate event: expect 202, 400 + REJECTED, 200 duplicate.
- Two identical events at the same instant: expect one incident (after F7).
- AI success with sample LOW/MEDIUM/HIGH/CRITICAL events; force AI failure: expect AI_FALLBACK and mandatory approval.
- Approval: approve, reject, wait for timeout, and break the email node: four distinct outcomes.
- Recovery of an approval-timeout incident: must not dispatch without approval (after F2).
- Dispatch success, Slack failure with simulate switch, self-healing on next tick, retries exhausted ⇒ ESCALATED.
- Run two Watchdog executions at once: one dispatch per incident (after F11).
- Force an error in each workflow: Slack alert and FAILED or audit-only as designed.
- Illegal transitions rejected by CT-02; audit row exists for every change.
- If CT-07 is enabled: note created once per final state; contact not found; HubSpot failure.

11. Delivered diagram files

Editable: **CT_Architecture_Blueprint.drawio** (6 pages). Static: SVG and PNG for each view. Mermaid equivalents of views 1, 2A, 2B and 3. The draw.io XML is well-formed and the SVG/PNG were rendered and inspected; the draw.io file was not opened in the draw.io application, so minor layout differences there are possible. Mermaid files rendered successfully.

12. Outstanding questions and assumptions

- SQL schema, constraints and indexes for all tables (none supplied).

- Is the Error Workflow configured outside the exported settings for CT-01/CT-05? The exports show it is not.
- Supabase key privilege level; HubSpot app scopes; Gemini model actually used.
- Whether workflows were ever executed end-to-end; execution history was not provided.
- Which Slack channel the hard-coded ID refers to, and who the intended approver is.
- Assumed: node defaults (Supabase matchType = any; Schedule = 5 min; chat memory table name) are as in current n8n; **Verify**.
- Assumed: the superseded CT-01 (4) and older CT-07 files are not deployed.